MEETING OR CONTACT REPORT

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Organization & Location: Los Alamos Scientific Laboratories Los Alamos, New Mexico	Date: October 17, 1958 Reported By: H. G. Kolsky
Project: STRETCH Mathematical Planning Group Meeting	Department: 749 Follow-up Date:

PERSONNEL PARTICIPATING:
(Place asterisk next to those on distribution list. Other distribution show at end of report)

LASL	IBM
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L. Gatt	
R. Lazarus	
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The main purpose of this informal meeting was to hear a progress report from IBM Poughkeepsie by Dr. Kolsky. The following topics were discussed briefly:

1. Present Status of the Data Flow Model.
2. Recent Instruction changes (augments, compare range, etc.)
3. Present estimates as to STRETCH system requirements for power, air conditioning, etc.
4. Description of the proposal to Westinghouse.
5. Description of 709-transistorized system.
6. The possibility of a real-time clock.
7. Performance Comparison of half microsecond and two microsecond memory.
8. Schedule of LASL test problem
9. Discussion of punch requirements, etc.
10. Miscellaneous questions concerning machine details.

H. G. Kolsky

1. The Data Flow Model:

The data flow model is constructed using standard frames and transistor cards. Its main purpose is for the engineers to test out the components and circuit logic under realistic conditions before freezing the final designs. Another important purpose is to measure the circuit speeds and get better estimates for ultimate arithmetic speeds.

The data flow model consists of two main parts: an 8 bit adder and a 4 by 4 bit multiplier giving an 8 bit product. These are built using the so-called "standard card" which bears 4 or 6 transistors plus other circuit elements. Later the adder will be rebuilt using the "double card" which holds up to 26 transistors plus other elements. This "double card" will have many indirect beneficial effects on the speed by making the units more compact.

At present, the adder has been running for about two weeks. It executes the path thru the adder in 0.45 usec. which is slightly less than the time originally estimated. The multiplier should be running within two weeks. There seems to be every indication that the floating point speeds given at the July meeting (i.e. floating add 0.95 us. floating mpy. 1.75 us, floating divide 7.0 us.) will be achieved and perhaps slightly bettered.

Not much can be said concerning reliability as yet since the apparatus is being worked on all the time. However, it has been left running all night several times without difficulty.

2. Recent Instruction Changes:

Two advance copies of the recent changes in the manual involving the augment, add to magnitude, and compare range instructions were given to Mr. Gatt.

The VFL binary multiply and divide were discussed briefly. Since these instructions are done in the floating point arithmetic unit and are restricted to 48 bits some speed gain may be had by returning the product to the accumulator at offset 20 rather than offset 0 as presently specified. The possible advantages and disadvantages of this were discussed briefly. LASL expressed no feelings pro or con concerning offset 20, being willing to leave this decision to IBM.

3. Present estimates concerning power, air conditioning, etc.:

Recently a letter was sent to the National Bureau of Standards giving them the following estimates concerning the physical requirements for a STRETCH system. This information was furnished to help them plan a new building. They are considered conservative estimates and include I/O equipment.

Room: 40 X 50 (Computer is assumed L-shaped, I/O equipment fits in the L)
Power: 50 Kilowatts
Air Conditioning: 25 tons
Floor Loading: 50 lbs/sq ft. approximately. (not over 100 lbs/sq ft. locally)

4. The STRETCH Proposal to Westinghouse:

On October 2, IBM made a proposal to enter into a contract for the purchase of a STRETCH computer system consisting of the following units:

One (1)	Central Processing Unit - including the indexing arithmetic unit, the computer control circuits, the arithmetic unit, the computer bus system, and the basic eight (8) channel exchange.
One (1)	Console
One (1)	Inquiry Unit
One (1)	Console and Inquiry Adapter
Six (6)	Magnetic Core Memory Units - each containing 16,384 words of 64 bits.
One (1)	High Speed Exchange
One (1)	Disk Control Unit
One (1)	High Speed Disk Unit - having a capacity of 1,048,576 words
One (1)	7500 Card Reader
One (1)	Card Reader Adapter
One (1)	7550 Card Punch
One (1)	Card Punch Adapter
One (1)	7400 Printer
One (1)	Printer Adapter
Two (2)	Tape Control Units
Eight (8)	729-IV Tape Units

The I/O units are those from the 7070 system. These were used to avoid specifying unannounced mechanical devices.

LASL was shown a copy of the system description which was given to Westinghouse-Bettis and GE-KAPL. They were also told the price (in confidence) and the date by which Westinghouse must reply.

5. The 709-Transistorized System:

The fact that IBM is considering transistorizing the 709 computer was announced at the SHARE meeting in early September. The transistorized 709 is identical in order structure and logic to the tube 709. The only major change is in the DSU channels. The DSU will be combined into the main frame and more channels will be available. The machine will be 5 times faster internally than the regular 709. The STRETCH 2 usec memories and STRETCH type transistors will be used in the circuits which enables the machine to operate on a 2.4 usec cycle instead of a 12 usec cycle.

The 709-Transistorized has been bid to BMEWS, and a commercial program will probably be launched in a few months.

6. The Real-Time Clock:

The fact that a real-time clock is being considered for STRETCH in addition to the elapsed time clock was mentioned. Both clocks would be counted by a 1024 cycle per second oscillator. The real time clock counts up while the elapsed time clock counts down. The real time clock is read-only and no interrupt occurs when it resets to zero. One complete cycle would take about 777 days of continuous running. LASL felt that such a clock would be very useful and it would be welcomed by them.

7. Performance comparison of half and two microsecond memories

Timing Simulation studies done in the past few months have come up with some interesting results concerning the effect on system performance of the half-microsecond memory. The following cases seem to be the important ones:

a. Half-microsecond memory used for data
    1. Short field data (data-fetch limited)
    2. Long field data (arithmetic limited)

b. Half-micrsoecond memory used for instructions
    1. Logic and branching codes (instruction-fetch limited)
    2. Straight computation codes (arithmetic limited)

The simulator runs have shown that replacing the half microsecond memory boxes by regular two microsecond memory makes no difference in the cases a (2) and b (2) where the problems are arithmetic limited.

In case a (1) the effect on the computer speed can be considerable. For example, in the case of a series of 8 bit adds, the computer speed is reduced as much as 37% if the data is stored in two microsecond memory instead of half microsecond memory. This type of operation is quite important for the HARVEST system.

In case b (1) for regular scientific problems the effect is present but much smaller. The average of 5 typical scientific problems showed the following speed reductions:

	Reduction in Computer Speed		
Instruction Memory	STANDARD	SIGMA	HARVEST
2 half-microsec memories	0	0	0
2 two-microsec memories	- 3.7%	-2.5%	-1.8%
no separate instruction memory	-15.9%	-6.4%	-3.8%

The worst case for Sigma was -8.8% for the Monte Carlo branching problem.

Considerations of this type lead to IBM's proposing 6 boxes of 2 us memory to Westinghouse rather than 2 boxes of half and 4 boxes of two microsecond memory.

There are also economic factors which make IBM question the advisibility of offering the half-microsecond memory as a standard commercial item. The cost per box is about the same, so the cost per word is 16 times greater for the half microsecond memory since it contains 1,024 words instead of 16,384.

8. Schedule of LASL Test Problem

IBM requested that LASL furnish a time schedule and preliminary description of the test problem mentioned in the contract.

At present LASL is considering using a number of smaller test problems tailored to test individual features rather than one large problem. Lazarus described the difficulties inherent in debugging a large program at the same time the computer is being debugged. The question of how these smaller test problems would overlap diagnostic programs was raised.

LASL plans to wait until STRAP is completely formulated before starting coding in earnest. A preliminary schedule and description will be sent to IBM in about a month.

9. Punch Requirements on STRETCH

The whole area of STRETCH I/O equipment is being reviewed at IBM. Some of the questions under discussion are: Is on-line card punching necessary on STRETCH? What are the main uses for punching? What is the "best" speed for a punch? How sensitive are the uses to punch speed?

In the discussion which followed it was generally agreed that on-line card punching volume will be quite small, restricted mainly to control cards and file reference cards. It was also agreed ,however, that these uses were very hard to accomplish any other way.

Another point made was that the distinction between "on-line" and "off-line" becomes a little obscure when one has a multiprogrammed card to tape operation going on at the same time another calculation is proceeding. Under these conditions the higher the card punching rate the better.

If there is only a slow punch (in particular, 100 per minute) on the machine, the volume punching would be put on off-line tape to card equipment. A high speed punch would encourage multiprogrammed "off-line" operation on the machine. The extra space and cost of the peripheral card equipment was discussed.

As background information: LASL has on order a peripheral tape to card machine, as well as a 500 line per minute tape to printer, and a Stromberg-Carlson tape to printer. They decided not to order a CRT mainly because of the film processing difficulties locally.

They presently have two 150 line per minute tape to printers which are running 24 hours a day most of the time. The reason for ordering the tape to card machine was not so much to speed up 704 operation as to reduce the impossible tape library situation which has developed. They have about 1500 tapes on file with more needed all the time. The only practical way to release these tapes is to provide for convenient card storage of the old tape information.

There has been a steady decrease in the amount of on-line punching on the 704's at LASL mainly because of machine speed considerations. Unfortunately this reduction in punching has generated the above tape file bottleneck.

10. Paper Tape on STRETCH

The above discussion on punches drifted to the subject of paper tape punches and readers. The well-known advantages of paper tape for typewriter input and output were mentioned again. The LASL feeling was that it was important to have paper tape facilities on STRETCH, but that paper tape did not substitute for card punches and readers.

11. Data Transmission

LASL is interested in hearing details concerning possible systems for data transmission to STRETCH from remote sites and the feasibility of operating the computer from such remote sites.

Weather calculations at the Nevada Test site is an example of this type of requirement.

12. Questions asked concerning machine details (to be transmitted to Engineering).

1. Why don't we have the complete set of transmit and Swap orders (full-half immediate-direct) instead of just some of them?

2. What about a "transmit block of bits" order?

3. Why the restruction against putting instructions in index registers?

4. In the indicator branch instructions Bit #29 being 0 for "leave", and 1 for "set to zero" seems backwards. Why is it this way?

5. "Branch Relative" should be called "skip relative" . What happens when one gives Branch Relative -1 (to self)?

6. What is one good use for the "set and wait" instruction?

7. When a load indirect, execute, or execute indirect contains a long chain, how does one investigate whether or not it is stuck without waiting for the elapsed time interrupt? If one is disabled in such a condition, how does one ever get out?

8. A general question: What indicators are set if one calls from, stores to, or tests non-existant words or bits? (For example, branch on bit #63.)

9. What provisions are being made for a break-point type operation? The high-order address bit was considered at one time. What is its present status? This is still considered important by LASL.

10. General question: When does address #8 differ from the accumulator? For example: In the following code

Load (exp = 20, Mantissa = M)
Unnormalized Floating Add (exp = 48, M = 0)
Store low order negative address 8

This should make the accumulator exp = 0, M = fractional part of M, reversed sign in sign byte register. Does it?

11. What happens on "Store Square Root negative" to address 8?

12. What about the possibility of leaving the square root in the accumulator as well as storing it?

13. Next Meeting

The date of the next meeting was discussed but not decided. The possibility of having a meeting in connection with the Joint Eastern Computer Conference in December was mentioned.

The recent meetings of LASL with Applied Programming concerning the assembly program, etc., have answered most questions which have arisen, so that a "full-dressed" meeting of the Mathematical Planning Group has not been necessary.

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